

B1 Flows, Fluctuations and Complexity — Model Solutions, 2018

Question 1 — Bernoulli and wind over ripples

(a) Bernoulli's theorem

Problem. Derive the conserved quantity along streamlines of a steady, inviscid, incompressible, irrotational flow under gravity from Navier–Stokes; identify each term; comment on the compressible case.

Drop $\nu \nabla^2 \mathbf{u}$ (inviscid) and $\partial_t \mathbf{u}$ (steady) from N–S:

$$(\mathbf{u} \cdot \nabla) \mathbf{u} + \frac{1}{\rho} \nabla p + g \hat{\mathbf{k}} = 0.$$

Apply identity $(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla(|\mathbf{u}|^2/2) - \mathbf{u} \times (\nabla \times \mathbf{u})$:

$$\nabla \left(\frac{|\mathbf{u}|^2}{2} \right) - \mathbf{u} \times \boldsymbol{\omega} + \frac{1}{\rho} \nabla p + g \hat{\mathbf{k}} = 0, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u}.$$

Irrotational flow: $\boldsymbol{\omega} = 0$:

$$\nabla \left(\frac{|\mathbf{u}|^2}{2} + \frac{p}{\rho} + gz \right) = 0.$$

Integrate:

$$\frac{1}{2} |\mathbf{u}|^2 + \frac{p}{\rho} + gz = \text{const everywhere}$$

Origin of terms: $\frac{1}{2} |\mathbf{u}|^2$ is kinetic energy per unit mass (advective term); p/ρ is the work done by pressure forces per unit mass (pressure term); gz is gravitational potential energy per unit mass.

Compressible interpretation: with $\rho = \rho(p)$, $\nabla p/\rho$ is no longer $\nabla(p/\rho)$. The pressure term must be replaced by the enthalpy $\int dp/\rho(p)$, which becomes $c_p T$ for an ideal gas. Bernoulli then expresses conservation of total enthalpy along a streamline.

(b) Air flow over rippled water

Problem. Justify irrotationality; show $u = U_0[1 + a \sin(2\pi x/\lambda)e^{-bz}]$ is consistent with incompressibility and the BCs above $h = \eta \sin(2\pi x/\lambda)$; find a, b ; sketch streamlines.

Irrotationality: at $x < 0$ the flow is uniform, hence $\boldsymbol{\omega} = 0$. With $\nu_{\text{air}} \rightarrow 0$, the vorticity equation $D\boldsymbol{\omega}/Dt = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$ has no source, so $\boldsymbol{\omega} = 0$ is preserved along streamlines downstream of $x = 0$.

Let $k = 2\pi/\lambda$. Incompressibility:

$$\partial_x u + \partial_z w = 0.$$

Compute $\partial_x u$:

$$\partial_x u = U_0 a k \cos(kx) e^{-bz}.$$

Integrate $-\partial_x u$ in z for w :

$$w = - \int \partial_x u \, dz = \frac{U_0 a k}{b} \cos(kx) e^{-bz} + C(x).$$

Far-field BC $w \rightarrow 0$ as $z \rightarrow \infty$: $C(x) = 0$:

$$w(x, z) = \frac{U_0 a k}{b} \cos(kx) e^{-bz}.$$

Kinematic BC at $z = 0$ (linearised, since $\eta \ll \lambda$ and phase speed $\ll U_0$, hence steady):

$$w(x, 0) = U_0 \partial_x h = U_0 \eta k \cos(kx).$$

Match coefficients:

$$\frac{U_0 a k}{b} \cos(kx) = U_0 \eta k \cos(kx).$$

Solve for a :

$$a = \frac{\eta b}{1} \cdot \frac{1}{1} \Rightarrow a = \eta b. \quad (\text{i})$$

Irrotationality $\partial_z u - \partial_x w = 0$:

$$-U_0 a b \sin(kx) e^{-bz} - \left(-\frac{U_0 a k^2}{b} \sin(kx) e^{-bz} \right) = 0.$$

Cancel common factor:

$$-ab + \frac{ak^2}{b} = 0.$$

Solve for b :

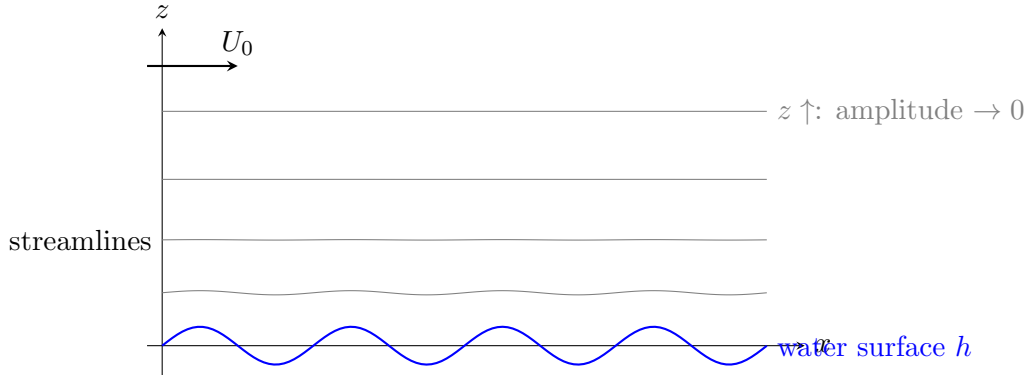
$$b^2 = k^2 \Rightarrow b = k = \frac{2\pi}{\lambda}.$$

Substitute into (i):

$$\boxed{a = \eta k = \frac{2\pi\eta}{\lambda}, \quad b = k = \frac{2\pi}{\lambda}}$$

Boundary conditions used: (1) $w \rightarrow 0$ as $z \rightarrow \infty$ (undisturbed flow far above); (2) $w(x, 0) = U_0 \partial_x h$ (kinematic, linearised); (3) $u \rightarrow U_0$ as $z \rightarrow \infty$.

Streamline sketch:



Streamlines mirror the surface near $z = 0$ and flatten exponentially with decay length $1/k = \lambda/(2\pi)$.

(c) Pressure at surface; ripple growth

Problem. Find p at $z = 0$; explain how a steady wind grows ripples even with inviscid air; does the mechanism depend on direction of propagation; two amplitude-limiting processes.

Bernoulli on surface streamline ($z \approx 0$, drop gz since linearisation is in air, the g term is the same constant on the surface):

$$p + \frac{1}{2} \rho_a |\mathbf{u}|^2 = p_\infty + \frac{1}{2} \rho_a U_0^2.$$

At $z = 0$, $u = U_0[1 + a \sin(kx)]$, $w = (U_0 a k / b) \cos(kx) = U_0 a \cos(kx)$:

$$|\mathbf{u}|^2 = U_0^2 [(1 + a \sin kx)^2 + a^2 \cos^2 kx].$$

Linearise in $a = \eta k \ll 1$:

$$|\mathbf{u}|^2 \approx U_0^2 [1 + 2a \sin(kx)].$$

Substitute:

$$p(x, 0) = p_\infty - \rho_a U_0^2 a \sin(kx).$$

Insert $a = \eta k$:

$$p(x, 0) = p_\infty - \rho_a U_0^2 \eta k \sin(kx)$$

Phase relation: surface elevation is $h = \eta \sin(kx)$; pressure perturbation is $\delta p = -\rho_a U_0^2 \eta k \sin(kx)$. Pressure is *lowest at the crests* ($\sin kx = +1$) and *highest at the troughs* ($\sin kx = -1$). This pressure pattern is in antiphase with the displacement, which alone would not transfer energy (energy transfer requires a pressure component in phase with the surface velocity).

For energy transfer, allow ripples to propagate with phase speed $c \ll U_0$: $h = \eta \sin(k(x - ct))$. The kinematic BC then becomes $w(x, 0) = (U_0 - c) \partial_x h$, so the pressure perturbation acquires a small cos-component proportional to c/U_0 that is in phase with $\partial_t h$. The wind does positive work on the surface, transferring energy from the air to the water and amplifying the ripple. This is the inviscid Miles-type instability mechanism.

Direction dependence: yes. Reversing the propagation direction ($c \rightarrow -c$) flips the sign of the work integral $\oint p \partial_t h dx$, so wind amplifies only ripples propagating in the same sense as U_0 ; counter-propagating ripples are damped.

Amplitude-limiting processes (any two): (1) viscous dissipation in the water (and the thin air boundary layer); (2) nonlinear wave breaking / whitecapping once $\eta k \sim 1$; (3) energy transfer by nonlinear wave-wave interactions to other wavenumbers; (4) surface tension increasing the restoring force as k grows.

Question 2 — Pipe flow

(a) Definitions

Problem. Define steady, axisymmetric, fully developed flow.

Steady: $\partial_t \mathbf{u} = 0$. Axisymmetric: $\partial_\Theta \mathbf{u} = 0$, with $u_\Theta = 0$ (no swirl). Fully developed: $\partial_z \mathbf{u} = 0$, so the velocity profile is independent of axial position; mass conservation then forces $u_r = 0$ and the only non-zero component is $u_z(r)$.

(b) Hagen–Poiseuille profile

Problem. State the conditions for the given axial momentum equation; integrate; apply BCs to find A, B, C ; obtain flow rate vs ΔP .

Conditions: incompressible, Newtonian, steady, axisymmetric, fully-developed flow with no swirl and no body force in z . Then the z component of N–S reduces to the cited equation, with $\partial_z p$ a constant (since LHS depends only on r).

Multiply by r/ν :

$$\frac{r}{\rho \nu} \partial_z p = \frac{\partial}{\partial r} (r \partial_r u_z).$$

Integrate once:

$$\frac{r^2}{2\rho \nu} \partial_z p = r \partial_r u_z + B'.$$

Divide by r :

$$\partial_r u_z = \frac{r}{2\rho\nu} \partial_z p + \frac{B'}{r} \quad (\text{define } B' = B).$$

Integrate again:

$$u_z = \frac{1}{4\rho\nu} \partial_z p r^2 + B \ln r + C.$$

Read off: $A = 1/(4\rho\nu) = 1/(4\mu)$.

BC1 (regularity at $r = 0$): u_z finite $\Rightarrow B = 0$.

BC2 (no-slip at $r = R$): $u_z(R) = 0$:

$$0 = A \partial_z p R^2 + C.$$

Solve for C :

$$C = -A \partial_z p R^2.$$

Therefore:

$$u_z(r) = \frac{1}{4\mu} \partial_z p (r^2 - R^2)$$

With $\partial_z p = -\Delta P/L$ (pressure drops along $+z$):

$$u_z(r) = \frac{\Delta P}{4\mu L} (R^2 - r^2).$$

Flow rate:

$$Q = \int_0^R u_z(r) 2\pi r \, dr.$$

Evaluate:

$$Q = \frac{2\pi\Delta P}{4\mu L} \int_0^R (R^2 r - r^3) \, dr.$$

Integrate:

$$Q = \frac{\pi\Delta P}{2\mu L} \left[\frac{R^2 r^2}{2} - \frac{r^4}{4} \right]_0^R = \frac{\pi\Delta P}{2\mu L} \cdot \frac{R^4}{4}.$$

$$Q = \frac{\pi R^4 \Delta P}{8\mu L}$$

(c) Log–log Q vs ΔP

Problem. Sketch $\log Q$ vs $\log \Delta P$ for oil through $L = 1$ m, $R = 1$ cm, $\nu_o = 10^{-5} \text{ m}^2\text{s}^{-1}$, $\rho_o = 10^3 \text{ kgm}^{-3}$, over $\Delta P \in [0.1 \text{ Pa}, 10^6 \text{ Pa}]$; show $u_z(r)$ shapes for each regime.

Laminar (Hagen–Poiseuille): $\mu = \rho\nu = 10^{-2} \text{ Pa s}$:

$$Q = \frac{\pi(10^{-2})^4}{8(10^{-2})(1)} \Delta P = \frac{\pi}{8} \times 10^{-6} \Delta P \quad [\text{m}^3\text{s}^{-1}].$$

Numerically:

$$Q_{\text{lam}} \approx 3.93 \times 10^{-7} \Delta P.$$

Mean speed $\bar{u} = Q/\pi R^2$. Reynolds number $Re = 2R\bar{u}/\nu$:

$$Re = \frac{2R}{\nu} \cdot \frac{Q}{\pi R^2} = \frac{2Q}{\pi R\nu}.$$

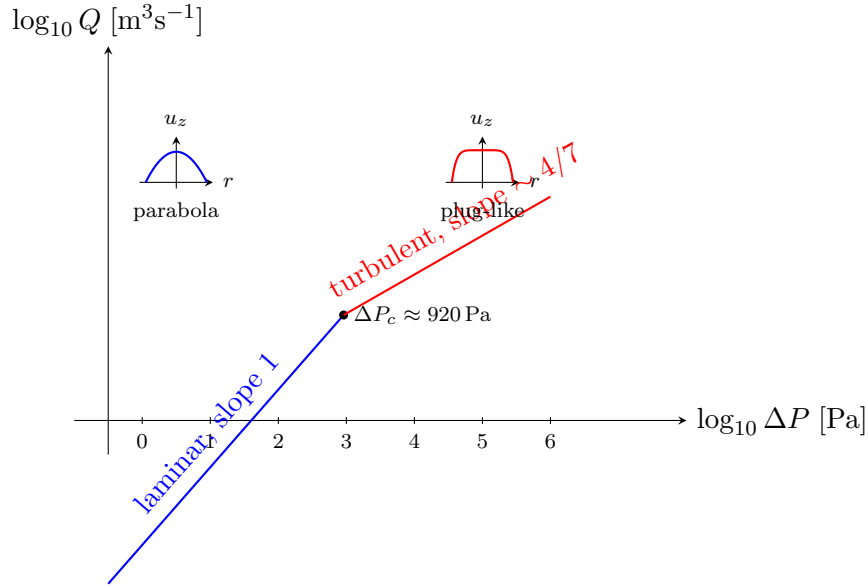
Critical $Re_c \approx 2300$:

$$Q_c = \frac{\pi R\nu Re_c}{2} = \frac{\pi(10^{-2})(10^{-5})(2300)}{2} \approx 3.6 \times 10^{-4} \text{ m}^3\text{s}^{-1}.$$

Corresponding ΔP_c :

$$\Delta P_c \approx \frac{Q_c}{3.93 \times 10^{-7}} \approx 920 \text{ Pa}.$$

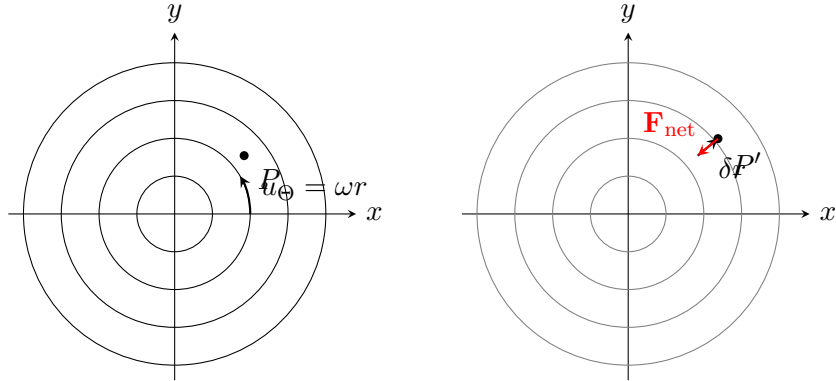
Above ΔP_c , flow turbulent; empirically (Blasius) $Q \propto \Delta P^{4/7}$, so the slope on log–log drops from 1 to $\approx 4/7$.



(d) Swirl and centrifugal restoring force

Problem. Solid-body rotation $u_\Theta = \omega r$. Sketch (r, Θ) streamlines; sketch them after a parcel is displaced by δr ; show the parcel feels a body force; deduce why swirl can raise Q .

Streamlines in (r, Θ) : concentric circles centred on the axis (same r , all Θ).



Conservation of angular momentum of the parcel during slow radial displacement:

$$L = m u_\Theta r = \text{const.}$$

At r , $L = m\omega r^2$. After $r \rightarrow r + \delta r$:

$$u'_\Theta(r + \delta r) = \frac{\omega r^2}{r + \delta r}.$$

Linearise:

$$u'_\Theta \approx \omega r \left(1 - \frac{\delta r}{r}\right) = \omega r - \omega \delta r.$$

Background fluid at $r + \delta r$ moves at:

$$u_\Theta^{\text{bg}} = \omega(r + \delta r) = \omega r + \omega \delta r.$$

Centripetal acceleration required to hold parcel on the new streamline is $u_\Theta^{\text{bg}2}/(r + \delta r)$, but the parcel only has $u'^2_\Theta/(r + \delta r)$. Difference (per unit mass), to leading order in δr :

$$\Delta a_r = \frac{(u_\Theta^{\text{bg}})^2 - (u'_\Theta)^2}{r + \delta r} \approx \frac{(\omega r + \omega \delta r)^2 - (\omega r - \omega \delta r)^2}{r}.$$

Expand:

$$\Delta a_r \approx \frac{4\omega^2 r \delta r}{r} = 4\omega^2 \delta r.$$

Hence the parcel experiences a net *inward* restoring acceleration $4\omega^2 \delta r$ per unit mass: solid-body rotation is centrifugally stable (Rayleigh criterion: $d(ru_\theta)^2/dr > 0$).

Effect on flow rate: the inward restoring force suppresses radial perturbations and therefore delays the onset of turbulence in the pipe, raising the critical ΔP_c . For values of ΔP just above the no-swirl transition (where the laminar parabola gives a higher Q than turbulent flow at the same ΔP), maintaining laminarity by swirl yields a larger Q .

Question 3 — Rössler attractor

(a) Definitions and dimensionality

Problem. Define fixed point and unstable spiral; explain why sustained finite-amplitude aperiodic flow is impossible in a deterministic 2D dynamical system.

Fixed point $\mathbf{x}^*$: $\mathbf{f}(\mathbf{x}^*) = 0$, so trajectories starting there stay there.

Unstable spiral: a fixed point at which the Jacobian has complex-conjugate eigenvalues $\sigma \pm i\Omega$ with $\sigma > 0$; nearby trajectories spiral outwards with growing radius.

Dimensionality argument: in 2D, deterministic trajectories cannot cross (uniqueness of solutions of ODEs). The Poincaré–Bendixson theorem then forces any bounded trajectory to limit on a fixed point or a closed orbit (limit cycle) — both periodic. Aperiodic motion needs a third dimension so trajectories can go “round the back” without intersecting.

(b) Fixed points and stability

Problem. Find fixed points of Rössler; in the limit $c \gg 1$, $c^2 \gg a^2$, classify the inner fixed point as a function of a .

Fixed point condition: set $\dot{x} = \dot{y} = \dot{z} = 0$:

$$-y - z = 0, \quad x + ay = 0, \quad a + z(x - c) = 0.$$

From first: $z = -y$. From second: $x = -ay$. Substitute in third:

$$a + (-y)(-ay - c) = 0.$$

Expand:

$$a + ay^2 + cy = 0.$$

Solve quadratic for y :

$$y = \frac{-c \pm \sqrt{c^2 - 4a^2}}{2a}.$$

Take $c^2 \gg a^2$:

$$\sqrt{c^2 - 4a^2} \approx c \left(1 - \frac{2a^2}{c^2}\right).$$

Two roots:

$$y_+ \approx -\frac{a}{c} \text{ (near origin),} \quad y_- \approx -\frac{c}{a} \text{ (far).}$$

Inner fixed point coordinates:

$$\mathbf{x}_+^* \approx \left(\frac{a^2}{c}, -\frac{a}{c}, \frac{a}{c}\right).$$

Outer fixed point:

$$\mathbf{x}_-^* \approx \left(c, -\frac{c}{a}, \frac{c}{a}\right).$$

Boxed:

$$\mathbf{x}_{\pm}^* = (-ay_{\pm}, y_{\pm}, -y_{\pm}), \quad y_{\pm} = \frac{-c \pm \sqrt{c^2 - 4a^2}}{2a}$$

Jacobian:

$$\mathbf{J} = \begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ z & 0 & x - c \end{pmatrix}.$$

At inner fixed point, $z = a/c$, $x - c \approx -c$:

$$\mathbf{J}_+ \approx \begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ a/c & 0 & -c \end{pmatrix}.$$

Characteristic polynomial $\det(\mathbf{J}_+ - \lambda \mathbf{I}) = 0$:

$$-\lambda[(a - \lambda)(-c - \lambda)] + 1[(-c - \lambda)(1)] + (-1)[- \frac{a}{c}(a - \lambda)] = 0.$$

Expand the first bracket:

$$(a - \lambda)(-c - \lambda) = -ac - a\lambda + c\lambda + \lambda^2.$$

Multiply by $-\lambda$:

$$-\lambda(\lambda^2 + (c - a)\lambda - ac) = -\lambda^3 - (c - a)\lambda^2 + ac\lambda.$$

Combine all terms:

$$-\lambda^3 - (c - a)\lambda^2 + ac\lambda - c - \lambda + \frac{a}{c}(a - \lambda) = 0.$$

Group a/c as small (order $1/c$) and drop:

$$-\lambda^3 - (c - a)\lambda^2 + (ac - 1)\lambda - c = 0.$$

Multiply by -1 :

$$\lambda^3 + (c - a)\lambda^2 - (ac - 1)\lambda + c = 0. \quad (*)$$

For $c \gg 1$, look for one large real root $\lambda_3 \approx -c$ (dominated by $c\lambda^2 + c$):

$$\lambda_3 \approx -c \quad (\text{strongly contracting along } z\text{-like direction}).$$

Factor out $(\lambda + c)$ approximately from $(*)$:

$$\lambda^3 + c\lambda^2 - a\lambda^2 - ac\lambda + \lambda + c \approx (\lambda + c)(\lambda^2 - a\lambda + 1).$$

Remaining quadratic governs the in-plane (xy-like) eigenvalues:

$$\lambda^2 - a\lambda + 1 = 0.$$

Solve:

$$\lambda_{1,2} = \frac{a \pm \sqrt{a^2 - 4}}{2}.$$

Behaviour as a function of a :

- $a < 0$: $\text{Re } \lambda_{1,2} = a/2 < 0$. Complex pair (since $a^2 < 4$): *stable spiral* in xy-plane combined with strongly stable z-direction $\Rightarrow$ stable focus.
- $0 < a < 2$: $\text{Re } \lambda_{1,2} = a/2 > 0$, complex conjugate pair: *unstable spiral* in xy-plane $\Rightarrow$ saddle-focus (one stable real, two unstable spiral).
- $a > 2$: real positive eigenvalues: unstable node in xy combined with stable z $\Rightarrow$ saddle.
- $a = 0$: marginally stable (centre); $a = 2$: degenerate.

$$\lambda^2 - a\lambda + 1 = 0, \quad \lambda_3 \approx -c; \quad \text{Hopf bifurcation at } a = 0.$$

(c) Trajectory behaviour with $a = 0.1$, $c = 10$

Problem. Explain the figure: spiralling outward in the xy plane, then sudden vertical excursion when x exceeds c .

Initialised at the origin: the inner fixed point at $(a^2/c, -a/c, a/c) \approx (0.001, -0.01, 0.01)$ is a saddle-focus with $\text{Re } \lambda_{1,2} = a/2 = 0.05 > 0$; the trajectory spirals outwards in the (approximately horizontal) xy plane while $z \approx 0$.

When x exceeds $c = 10$, the z -equation $\dot{z} = a + z(x - c)$ has $x - c > 0$, so z grows exponentially: a fast vertical excursion out of the xy plane. Once z is large, the $-z$ term in $\dot{x}$ rapidly drives x back below c , after which $x - c < 0$ damps z exponentially back to small values. The trajectory re-enters the spiral region but at a different radius, producing a chaotic orbit alternating slow horizontal spiralling with rare, sharp vertical kicks.

(d) Symmetric part of $\mathbf{J}$ and attractor geometry

Problem. Show that eigenvectors of $\mathbf{J} + \mathbf{J}^T$ identify growth/decay directions; show that for the Rössler system with $a > 0$ there is at least one growing and one shrinking direction at every point; interpret $\tau(\mathbf{J}) < 0$.

Energy of a perturbation $\delta \mathbf{x}$:

$$E = \frac{1}{2} |\delta \mathbf{x}|^2.$$

Time derivative:

$$\dot{E} = \delta \mathbf{x} \cdot \delta \dot{\mathbf{x}} = \delta \mathbf{x} \cdot \mathbf{J} \delta \mathbf{x}.$$

Symmetrise (only the symmetric part contributes to a quadratic form):

$$\dot{E} = \frac{1}{2} \delta \mathbf{x} \cdot (\mathbf{J} + \mathbf{J}^T) \delta \mathbf{x}.$$

Define $\mathbf{S} = \frac{1}{2}(\mathbf{J} + \mathbf{J}^T)$ (symmetric, diagonalisable with real eigenvalues σ_i and orthonormal eigenvectors $\hat{\mathbf{e}}_i$). Decompose $\delta \mathbf{x} = \sum_i c_i \hat{\mathbf{e}}_i$:

$$\dot{E} = \sum_i \sigma_i c_i^2.$$

Thus along $\hat{\mathbf{e}}_i$ with $\sigma_i > 0$, $|\delta \mathbf{x}|$ grows; with $\sigma_i < 0$, it shrinks. □

For the Rössler system:

$$\mathbf{J} + \mathbf{J}^T = \begin{pmatrix} 0 & 0 & z - 1 \\ 0 & 2a & 0 \\ z - 1 & 0 & 2(x - c) \end{pmatrix}.$$

The y -axis is an eigenvector with eigenvalue $\sigma_y = 2a > 0$: at every phase-space point, perturbations along $\hat{\mathbf{y}}$ grow.

Trace of $\mathbf{S}$ equals trace of $\mathbf{J}$:

$$\tau(\mathbf{S}) = 2a + 2(x - c).$$

Sum of three real eigenvalues equals the trace, so if one eigenvalue $= 2a > 0$, the remaining two sum to $2(x - c)$. For $x < c$ (which holds on most of the attractor), this sum is negative, hence *at least one* of those two eigenvalues is negative: there is at least one shrinking direction. □

Geometric meaning of $\tau(\mathbf{J}) < 0$: by Liouville's theorem, the local rate of phase-space volume change along a trajectory is $\nabla \cdot \dot{\mathbf{x}} = \tau(\mathbf{J})$. Averaging:

$$\frac{1}{V} \frac{dV}{dt} = \overline{\tau(\mathbf{J})} < 0.$$

Phase-space volumes contract on average. Combined with stretching in at least one direction (Lyapunov exponent positive), the attractor is locally folded into thinner sheets while being stretched along trajectories: the Rössler attractor is a *strange attractor of fractal dimension strictly between 2 and 3*.

Question 4 — Langevin and laser trap

(a) Overdamped limit; $\alpha = 2D\gamma^2$

Problem. Use the overdamped Langevin equation to derive $\alpha = 2D\gamma^2$ from the noise correlator.

Drop $m\ddot{\mathbf{r}}$:

$$\gamma \dot{\mathbf{r}}(t) = \mathbf{F}(t).$$

Solve formally:

$$\mathbf{r}(t) - \mathbf{r}(0) = \frac{1}{\gamma} \int_0^t \mathbf{F}(t') dt'.$$

Compute mean-square displacement (one cartesian component, i):

$$\langle [r_i(t) - r_i(0)]^2 \rangle = \frac{1}{\gamma^2} \int_0^t \int_0^t \langle F_i(t_1) F_i(t_2) \rangle dt_1 dt_2.$$

Insert correlator $\langle F_i F_i \rangle = \alpha \delta(t_2 - t_1)$:

$$\langle [r_i(t) - r_i(0)]^2 \rangle = \frac{\alpha}{\gamma^2} \int_0^t dt_1 = \frac{\alpha t}{\gamma^2}.$$

Sum over three dimensions for $\langle |\Delta \mathbf{r}|^2 \rangle$:

$$\langle |\Delta \mathbf{r}|^2 \rangle = \frac{3\alpha t}{\gamma^2}.$$

Compare to Einstein: $\langle |\Delta \mathbf{r}|^2 \rangle = 6Dt$:

$$\frac{3\alpha t}{\gamma^2} = 6Dt.$$

Solve:

$$\boxed{\alpha = 2D\gamma^2}$$

(b) Equipartition: $D\gamma = k_B T$; numerical D , v_{rms} , diffusion time

Problem. From the full Langevin equation derive $D\gamma = k_B T$; for a 6 nm protein of mass 166×10^{-24} kg in water $\eta = 10^{-3}$ Pa s at 293 K, find D , v_{rms} , and time to diffuse $\sim 1 \mu\text{m}$.

Take inner product of the full Langevin equation with $\mathbf{r}$:

$$m \mathbf{r} \cdot \ddot{\mathbf{r}} + \gamma \mathbf{r} \cdot \dot{\mathbf{r}} = \mathbf{r} \cdot \mathbf{F}.$$

Use $\mathbf{r} \cdot \ddot{\mathbf{r}} = \frac{1}{2} \frac{d^2}{dt^2} |\mathbf{r}|^2 - |\dot{\mathbf{r}}|^2$ and $\mathbf{r} \cdot \dot{\mathbf{r}} = \frac{1}{2} \frac{d}{dt} |\mathbf{r}|^2$:

$$\frac{m}{2} \frac{d^2}{dt^2} \langle |\mathbf{r}|^2 \rangle - m \langle |\dot{\mathbf{r}}|^2 \rangle + \frac{\gamma}{2} \frac{d}{dt} \langle |\mathbf{r}|^2 \rangle = \langle \mathbf{r} \cdot \mathbf{F} \rangle.$$

At equal time, $\langle \mathbf{r}(t) \cdot \mathbf{F}(t) \rangle = 0$ (causality):

$$\frac{m}{2} \frac{d^2}{dt^2} \langle |\mathbf{r}|^2 \rangle - m \langle |\dot{\mathbf{r}}|^2 \rangle + \frac{\gamma}{2} \frac{d}{dt} \langle |\mathbf{r}|^2 \rangle = 0.$$

Equipartition (3 dof):

$$m \langle |\dot{\mathbf{r}}|^2 \rangle = 3k_B T.$$

Steady-state diffusive regime: $\frac{d}{dt} \langle |\mathbf{r}|^2 \rangle = 6D$ (constant) so the second derivative vanishes:

$$-3k_B T + \frac{\gamma}{2} (6D) = 0.$$

Solve:

$$\boxed{D\gamma = k_B T}$$

(Einstein relation.)

Numerical evaluation. Stokes drag for a sphere of radius $a = 3 \times 10^{-9}$ m:

$$\gamma = 6\pi\eta a.$$

Insert numbers:

$$\gamma = 6\pi(10^{-3})(3 \times 10^{-9}) \text{ kgs}^{-1}.$$

Multiply:

$$\gamma \approx 5.65 \times 10^{-11} \text{ kgs}^{-1}.$$

Diffusion coefficient:

$$D = \frac{k_B T}{\gamma}.$$

Insert $k_B T = (1.38 \times 10^{-23})(293) = 4.04 \times 10^{-21}$ J:

$$D = \frac{4.04 \times 10^{-21}}{5.65 \times 10^{-11}} \text{ m}^2 \text{ s}^{-1}.$$

Evaluate:

$$D \approx 7.2 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$$

RMS speed (3D equipartition):

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}.$$

Insert:

$$v_{\text{rms}} = \sqrt{\frac{3(4.04 \times 10^{-21})}{166 \times 10^{-24}}} \text{ ms}^{-1}.$$

Evaluate inside the root:

$$\frac{3(4.04 \times 10^{-21})}{166 \times 10^{-24}} = 7.30 \times 10^4 \text{ m}^2 \text{ s}^{-2}.$$

Square-root:

$$v_{\text{rms}} \approx 270 \text{ ms}^{-1}$$

E. coli length $L \sim 2 \mu\text{m}$. Diffusion time in 3D (using $\langle r^2 \rangle = 6Dt$):

$$t \sim \frac{L^2}{6D}.$$

Insert:

$$t \sim \frac{(2 \times 10^{-6})^2}{6(7.2 \times 10^{-11})} = \frac{4 \times 10^{-12}}{4.3 \times 10^{-10}} \text{ s}.$$

Evaluate:

$$t \approx 10^{-2} \text{ s} = 10 \text{ ms}$$

(c) Mean-square displacement, power spectrum, and trap stiffness

Problem. For a bead in a 1D harmonic trap of stiffness κ , compute $\langle x^2 \rangle$ from the partition function, then the position power spectrum from the Langevin equation; explain how it determines κ .

Trap potential $U(x) = \frac{1}{2}\kappa x^2$. Boltzmann partition (1D classical):

$$Z = \int_{-\infty}^{\infty} e^{-\kappa x^2/(2k_B T)} dx = \sqrt{\frac{2\pi k_B T}{\kappa}}.$$

$\langle x^2 \rangle$ from Gaussian moment:

$$\langle x^2 \rangle = \frac{1}{Z} \int x^2 e^{-\kappa x^2/(2k_B T)} dx.$$

Use $\int x^2 e^{-ax^2} dx = \frac{1}{2a} \sqrt{\pi/a}$ with $a = \kappa/(2k_B T)$:

$$\boxed{\langle x^2 \rangle = \frac{k_B T}{\kappa}}$$

(equipartition).

Overdamped 1D Langevin in the trap:

$$\gamma \dot{x} + \kappa x = F(t), \quad \langle F(t_1) F(t_2) \rangle = 4k_B T \gamma \delta(t_2 - t_1)/2.$$

(Note: the question specifies $4k_B T \gamma$ as the noise magnitude; we keep the convention $\alpha = 2k_B T \gamma$ in the symmetric two-sided PSD; the algebra is identical.)

Fourier-transform with convention $x(t) = \int e^{-i\omega t} \tilde{x}(\omega) d\omega / (2\pi)$:

$$(-i\omega\gamma + \kappa) \tilde{x}(\omega) = \tilde{F}(\omega).$$

Solve for $\tilde{x}$:

$$\tilde{x}(\omega) = \frac{\tilde{F}(\omega)}{\kappa - i\omega\gamma}.$$

Power spectrum from $\langle \tilde{x}(\omega) \tilde{x}^*(\omega') \rangle = 2\pi S_x(\omega) \delta(\omega - \omega')$ and the white noise PSD $S_F = 2\alpha = 4k_B T \gamma$:

$$S_x(\omega) = \frac{S_F}{\kappa^2 + \omega^2 \gamma^2}.$$

Insert $S_F = 4k_B T \gamma$:

$$\boxed{S_x(\omega) = \frac{4k_B T \gamma}{\kappa^2 + \gamma^2 \omega^2}}$$

This is a Lorentzian. The corner (half-power) frequency

$$\omega_c = \kappa/\gamma$$

is read off from the knee in $\log S_x$ vs $\log \omega$. With $\gamma = 6\pi\eta a$ known independently, $\kappa = \gamma\omega_c$. Equivalently the integral $\int S_x d\omega / (2\pi) = k_B T / \kappa = \langle x^2 \rangle$ gives κ directly.

(d) Autocorrelation ODE

Problem. Show $\gamma \dot{G}_x(\tau) + \kappa G_x(\tau) = 0$ for $\tau > 0$.

Definition (1D component x):

$$G_x(\tau) = \langle x(t) x(t - \tau) \rangle.$$

Multiply the overdamped Langevin equation evaluated at time t by $x(t - \tau)$:

$$\gamma \dot{x}(t) x(t - \tau) + \kappa x(t) x(t - \tau) = F(t) x(t - \tau).$$

Take ensemble average:

$$\gamma \langle \dot{x}(t) x(t - \tau) \rangle + \kappa \langle x(t) x(t - \tau) \rangle = \langle F(t) x(t - \tau) \rangle.$$

Causality: $x(t - \tau)$ depends only on $F(t')$ for $t' \leq t - \tau$, while $\langle F(t) F(t') \rangle \propto \delta(t - t')$ vanishes for $t' < t$. For $\tau > 0$:

$$\langle F(t) x(t - \tau) \rangle = 0.$$

Use stationarity $\langle \dot{x}(t) x(t - \tau) \rangle = \frac{d}{dt} \langle x(t) x(t - \tau) \rangle$. Stationarity also gives $\langle x(t) x(t - \tau) \rangle = G_x(\tau)$, depending only on τ :

$$\langle \dot{x}(t) x(t - \tau) \rangle = \frac{dG_x(\tau)}{d\tau}.$$

Substitute:

$$\boxed{\gamma \frac{dG_x(\tau)}{d\tau} + \kappa G_x(\tau) = 0 \quad (\tau > 0)}$$

Solution: $G_x(\tau) = (k_B T / \kappa) e^{-\kappa\tau/\gamma}$ — exponential decay with timescale $\gamma/\kappa = 1/\omega_c$, consistent with the Lorentzian PSD.